H Likelihood of preference dataset

When we evaluate the likelihood of a preference dataset given a preference model (under the assumption that it follows a Boltzmann distribution), we seek to evaluate which class of preference model can better express the human data, given equivalent versions of the ground-truth reward function that was taught to human subjects prior to preference elicitation. Specifically, we note that reward functions that differ only by a constant scaling factor are equivalent under most definitions—including how they order policies given a start state distribution and, by consequence, their sets of optimal policies—and different scalings of the same ground-truth reward function are considered an equivalence class. Concretely, to evaluate the likelihood of a dataset given a preference model class and this equivalence class that includes the ground-truth reward function, we use the highest likelihood across a predefined list of positive scaling parameters, each of which multiplies the output of the ground-truth reward function. This scaling parameter can also be seen as scaling the difference in the two segment statistics and it therefore affects entropy of the probabilities given to preferring each segment, pushing them closer to 0.5 or to 0 and 1. Mathematically, it has the same effect as using a Boltzmann temperature parameter, making such a temperature parameter redundant in most settings and therefore not part of our standard description of the preference models. The predefined list of scaling parameters was chosen to cover the space in which these preference models have relatively high likelihoods. Alternatively, this scaling parameter could be learned via gradient descent and tested on a heldout set, like in k-fold cross validation, but we decided against this approach out of concerns that learning a scaling parameter is not representative of actually learning a reward function, where the reward function has unknown parameters beyond its scale.

For each preference dataset resulting from each experimental condition, we evaluate how well P_{regret} and P_{Σ_r} predict the dataset. We explore a range of possible reward scaling parameters for P_{regret} and P_{Σ_r}, computing the mean cross-entropy loss for each parameter and model over the dataset. The reward scaling parameters were selected to be exponentially spaced between approximately 1 and -1. The n-th reward scaling parameter is given by $p_n = ar^{n-1}$. We used 25 reward scaling parameters: the first 12 were generated with $a = 0.01$ and $r = 1.236$, the next 12 with $a = -0.01$ and $r = 1.236$, and the final parameter was set to 0.

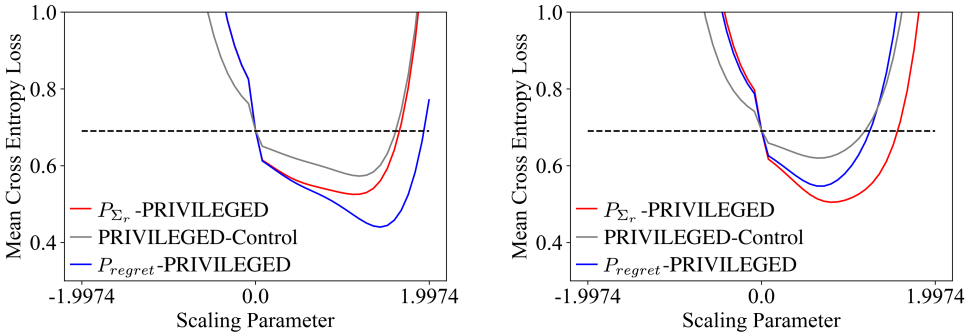

Figure 20: The mean cross entropy loss over each preference dataset resulting from the PRIVILEGED experiment (see Section 5.1) for all scaling parameters given the regret preference model (Left) and the partial return preference model (Right). If the loss is **lower** for the dataset of preferences influenced towards the target preference model than the control condition's dataset at a specific scaling parameter, it means the former is better predicted by—and more likely under—the target preference model given that scaling parameter. The regret preference model achieves the lowest mean cross-entropy loss over—and is therefore most predictive of—the P_{regret}-TRAINED dataset for all scaling parameters greater than 0. Similarly, the partial return preference model best predicts the P_{Σ_r}-TRAINED dataset for all scaling parameters greater than 0. This supports our hypothesis that showing subjects PRIVILEGED information about each segment's regret or partial return during preference elicitation does influence their preferences towards a specific preference model.

The plotted losses for each reward scaling parameter are illustrated in Figure 20 for the preference datasets obtained from the PRIVILEGED experiment, Figure 22 for the TRAINED experiment, and Figure 21 for the QUESTION experiment. Note that when fitting the regret preference model to a preference dataset for these plots, we apply the scaling parameter to the negated regret of a segment for easier visual comparison. These plots extend the results shown in Figures 5, 7, and 9. One incorrect conclusion to draw from those figures is that the proposed interventions simply train humans to better understand the ground-truth reward function, rather than to follow a specific preference model. We acknowledge the possibility of entangled effects relating to learning more about the ground-truth reward function rather than a specific preference model. However, we expect that in our experiments such effects are relatively minimal. Firstly, all human subjects already had a good understanding of the ground-truth reward function; we employed a comprehension test to filter out subjects who did not (see Appendix C.4). Further, in Figures 5, 7, and 9, we see that for all experiments the

loss over a condition's dataset is lower under the target preference model than all other conditions datasets under the same preference model. Had one condition trivially resulted in subjects better understanding the ground-truth reward function rather than the target preference model, we would expect to see that condition's dataset induce the lowest loss under either preference model. Figures 5, 7, and 9 illustrate that this is not the case.

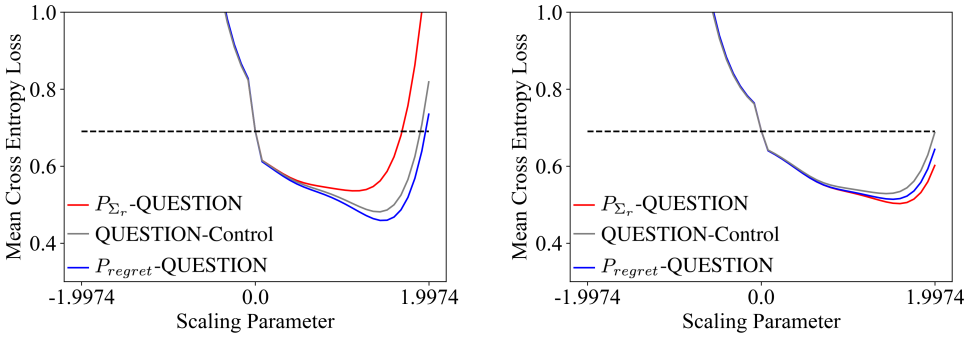

Figure 21: The mean cross entropy loss over each preference dataset resulting from the QUESTION experiment (see Section 5.3) for all scaling parameters given the regret preference model (Left) and the partial return preference model (Right). See Figure 20 for details on how to interpret this graph. The regret preference model achieves the lowest mean cross-entropy loss over the P_{regret}-QUESTION dataset, closely followed by the QUESTION-Control dataset, for all scaling parameters greater than 0. The partial return preference model best predicts the P_{Σ_r}-QUESTION condition's dataset, closely followed by the P_{regret}-QUESTION dataset and then the QUESTION-Control dataset for all scaling parameters greater than 0. These results suggest that changing the preference elicitation QUESTION to influence preferences towards the regret preference model may not be effective, though the loss over the QUESTION-Control dataset given the regret preference model is already relatively low. Modifying the QUESTION to guide preferences towards the partial return preference model has a moderate effect on the datasets conformity to the preference model.

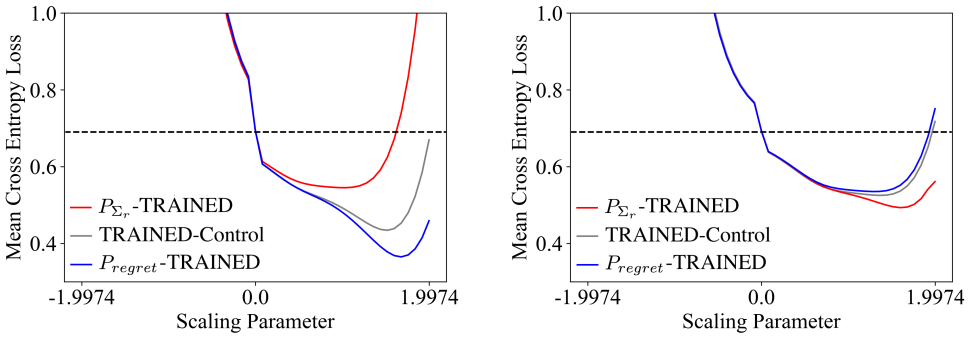

Figure 22: The mean cross entropy loss over each preference dataset resulting from the TRAINED experiment (see Section 5.2) for all scaling parameters given the regret preference model (Left) and the partial return preference model (Right). See Figure 20 for details on how to interpret this graph. For scaling parameters greater than 0, the regret preference model achieves the lowest mean cross-entropy loss over—and is therefore most predictive of—the P_{regret}-TRAINED dataset, followed by the TRAINED-Control dataset, and finally the P_{Σ_r}-TRAINED dataset. Similarly, the partial return preference model best predicts the P_{Σ_r}-TRAINED dataset followed by the P_{regret}-TRAINED and TRAINED-Control condition's datasets. This supports our hypothesis that teaching subjects about a specific preference model does influence their preferences towards that model.

We further characterize whether there is a difference in the likelihood of the control condition's dataset and the likelihood of the dataset that arises from influencing humans towards a specific preference model. The dataset of preferences in the PRIVILEGED experiment is unpaired so we perform a Mann-Whitney U test, while the dataset of preferences in the TRAINED and QUESTION experiments are paired so we perform a Wilcoxon paired signed-rank test. See Appendix C.2 for more details on how these datasets were constructed. All statistical tests are applied between the likelihoods over each dataset that result from the best scaling parameter—meaning the scaling parameter that induces the lowest mean cross-entropy loss.

Performing the Mann-Whitney U test between P_{Σ_r}-PRIVILEGED and PRIVILEGED-Control datasets results in $U = 1418607.0$, $p < 0.01$, and between the P_{regret}-PRIVILEGED and PRIVILEGED-Control datasets results in $U =$

1643392.5, $p < 0.01$. In the TRAINED and QUESTION experiments, to ensure that each condition contains the same segment pairs, we removed a segment pair from all conditions if any subject selected "Can't Tell" instead of indicating a preference for that pair. Performing the Wilcoxon test between the P_{Σ_r}-TRAINED and TRAINED-Control datasets results in $W = 6366.0$, $p < 0.01$, and between the P_{regret}-TRAINED and TRAINED-Control datasets results in $W = 12083.0$, $p < 0.01$. Additionally, performing the Wilcoxon test between the P_{Σ_r}-QUESTION and QUESTION-Control datasets results in $W = 7321.0$, $p < 0.05$—which we consider statistically significant—and between the P_{regret}-QUESTION and QUESTION-Control datasets results in $W = 2217.5$, $p = 0.685$—which we do not consider statistically significant.

I Accuracy over preference dataset

Computing the likelihood of a preference dataset given a preference model is an informative measure of how well that preference model describes the dataset. But, computing that likelihood also requires preference model P—defined in Equation 2 for partial return and Equation 4 for regret—which rests on the assumption that humans are Boltzmann-rational as instantiated via the logistic function. Therefore, to circumvent this assumption, we also compute the accuracy of the *noiseless* version of a given preference model over each dataset. These results are detailed below for all experiments.

We test the significance of these results using the Fisher's exact test (Upton, 1992). When executing the Fisher's exact test, for each condition in each experiment we construct a 2x2 contingency table where the first row is the number of preferences the noiseless target preference model classified correctly, the second row is the number of preferences the noiseless target preference model classified incorrectly, the first column is the dataset where subjects are influenced towards the target preference model, and the second column is the control condition's dataset.

PRIVILEGED Experiment Table 4 shows that, consistent with the results in Section 5.1, the noiseless version of the target preference preference model achieves higher accuracy on the preference dataset influenced toward the target model compared to the control condition dataset.

We conduct a Fisher exact test (Upton, 1992) to determine whether there is a significant difference in the proportion of preferences that the noiseless target preference model correctly classifies between the influenced preference dataset and the control condition dataset. We find a p-value of less than 0.001 when comparing the P_{Σ_r}-PRIVILEGED dataset to the PRIVILEGED-Control dataset, as well as when comparing the P_{regret}-PRIVILEGED dataset to the PRIVILEGED-Control dataset, indicating statistical significance.

TRAINED Experiment Table 5 presents the accuracy of the noiseless target preference model when predicting the P_{Σ_r}-TRAINED and P_{regret}-TRAINED datasets compared to the TRAINED-Control dataset. The accuracy over both the P_{Σ_r}-TRAINED and P_{regret}-TRAINED datasets is notably higher than the accuracy over the TRAINED-Control dataset given the respective preference model, supporting the results in Section 5.2.

Table 4: The accuracy of each condition's preference dataset from the PRIVILEGED experiment with respect to the *noiseless* version of the target preference model. If the accuracy is **higher** for the dataset of preferences influenced towards the target preference model than the control condition's dataset, it means the former is better predicted by the noiseless version of the target preference model.

Condition	Noiseless-P_{Σ_r} Accuracy	Noiseless-P_{regret} Accuracy
Control Condition	48.6%	55.9%
Influenced Towards Target Model	75.3%	75.8%

We conduct the Fisher exact test over the proportion of preferences predicted correctly by the noiseless target preference model; we find a p-value of 0.0012 when comparing the P_{Σ_r}-TRAINED dataset to the TRAINED-Control dataset, and of 0.0025 when comparing the P_{regret}-TRAINED dataset to the TRAINED-Control dataset, indicating statistical significance.

QUESTION Experiment As shown in Table 6, the accuracy of the noiseless partial return preference model is higher over the P_{Σ_r}-QUESTION dataset than the QUESTION-Control dataset. This indicates that changing the preference elicitation instruction to influence preferences towards partial return results in a preference dataset that is better predicted by partial return. The P_{regret}-QUESTION dataset, on the other hand, is not better predicted by regret than the control condition.

We conduct Fisher's exact test and find a p-value of 0.0157 when comparing the proportion of preferences predicted correctly by the noiseless partial return preference model for the P_{Σ_r}-TRAINED dataset versus the TRAINED-Control dataset, indicating statistical significance. We do not find a statistically-significant p-value when comparing the P_{regret}-TRAINED dataset to the TRAINED-Control dataset ($p = 0.7144$).